

Linear Algebra Review I: Determinants, Rank, Linear Systems, and More

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§1.1 Cofactor orthogonality

For an $n \times n$ matrix

$$A = (a_{ij})_{n \times n},$$

let A_{ij} denote the cofactor of a_{ij} . The page starts with the standard cofactor identity:

$$\sum_{k=1}^n a_{ik} A_{jk} = \begin{cases} |A|, & i = j, \\ 0, & i \neq j, \end{cases}$$

and similarly

$$\sum_{k=1}^n a_{ki} A_{kj} = \begin{cases} |A|, & i = j, \\ 0, & i \neq j. \end{cases}$$

When $i = j$, this is exactly Laplace expansion along the i -th row or column. When $i \neq j$, the sum is the determinant of a matrix with two equal rows or two equal columns, hence it is zero. This is the conceptual reason the adjugate identity

$$A A^* = A^* A = |A| I$$

holds.

§1.2 Vandermonde determinant

The note then proves the Vandermonde determinant:

$$D_n(x_1, x_2, \dots, x_n) = \begin{vmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \\ x_1^2 & x_2^2 & \dots & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \dots & x_n^{n-1} \end{vmatrix} = \prod_{1 \leq i < j \leq n} (x_j - x_i).$$

The proof strategy shown on the page is the usual elimination proof. Starting from the last row, subtract suitable multiples of the previous row so that each column except the first acquires a factor $x_j - x_1$. After pulling these factors out, the remaining determinant is a Vandermonde determinant of order $n - 1$ in $x_2, \dots, x_n$. Thus

$$D_n = \prod_{j=2}^n (x_j - x_1) D_{n-1}(x_2, \dots, x_n),$$

and induction gives the formula.

The page emphasizes that this formula may be used directly later. The practical meaning is that a determinant built from powers of variables often hides a product of pairwise differences.

§1.3 Block triangular determinants

For a block matrix of the form

$$\begin{vmatrix} A & O \\ * & B \end{vmatrix},$$

where A is $n \times n$ and B is $m \times m$, the determinant is

$$\begin{vmatrix} A & O \\ * & B \end{vmatrix} = |A| |B|.$$

Similarly,

$$\begin{vmatrix} A & * \\ O & B \end{vmatrix} = |A| |B|.$$

But when the zero block is on the other diagonal position, one must account for row or column swaps:

$$\begin{vmatrix} O & A \\ B & * \end{vmatrix} = (-1)^{mn} |A| |B|, \quad \begin{vmatrix} * & A \\ B & O \end{vmatrix} = (-1)^{mn} |A| |B|.$$

The handwritten comment warns that one cannot simply write every block matrix determinant as $|A||B|$. The location of the zero block and the parity of the swap matter.

§1.4 General Laplace expansion

A more general Laplace expansion is recorded. Choose k rows with indices $i_1, \dots, i_k$. Then

$$|A| = \sum_{1 \leq j_1 < \dots < j_k \leq n} (-1)^{i_1 + \dots + i_k + j_1 + \dots + j_k} MN,$$

where M is the k -minor formed by the selected rows and columns $j_1, \dots, j_k$, and N is the complementary cofactor determinant obtained after deleting those rows and columns.

The page notes that the case $k = 1$ is the usual row or column expansion. The formula is useful when many entries in a group of rows or columns are zero.

§1.5 Determinant splitting

One green block is labelled as splitting a row or column. The principle is multilinearity. If one row is a sum of two rows, then the determinant is the sum of two determinants. For example,

$$R_i = u + v \implies D(\dots, R_i, \dots) = D(\dots, u, \dots) + D(\dots, v, \dots).$$

The same is true for columns.

The example on the page uses a tridiagonal-looking determinant and splits the first row into a simple row plus a row containing the parameter a . The resulting recurrence has the form

$$D_n = a^n + aD_{n-1},$$

and iteration gives a closed expression. The lesson is that splitting is useful when one part becomes triangular and the other part reduces the order.

§1.6 Adding rows or columns

Another determinant trick is to add multiples of one row or column to another. This does not change the determinant. The note uses this to transform matrices into Vandermonde-like form.

A typical pattern is

$$C_j \leftarrow C_j + C_1$$

or

$$R_i \leftarrow R_i + \lambda R_j.$$

If the operation produces a row or column of ones or zeros, the determinant becomes easier to expand.

The handwritten margin says that this method is especially useful for symmetric or nearly Vandermonde matrices: add all rows, add all columns, then expand along the newly simplified row or column.

§1.7 Augmenting to a Vandermonde determinant

One page expands a determinant by adding an extra column containing a new variable y . The determinant becomes a Vandermonde determinant in

$$x_1, x_2, \dots, x_n, y.$$

After evaluating the enlarged determinant as

$$\prod_{1 \leq i < j \leq n} (x_j - x_i) \prod_{k=1}^n (y - x_k),$$

one compares the coefficient of a suitable power of y . This coefficient comparison recovers the original determinant.

The important technique is coefficient comparison after embedding the target determinant into a known determinant. This is a common trick for determinants containing missing powers or sums of variables.

§1.8 Row and column transformations

The scanned pages collect several row/column patterns:

- (i) proportional differences, where adjacent columns are subtracted to create repeated columns or zeros;
- (ii) proportional additions, where columns are successively modified by adding multiples from one side;
- (iii) total row or column addition, where all columns are added to one column to reveal a common factor;
- (iv) paired addition, where the upper half of rows is added to the lower half and the right half of columns is subtracted from the left half.

For example, a determinant with rows

$$a^2, (a+1)^2, (a+2)^2, (a+3)^2$$

for several values of a can be reduced by subtracting adjacent columns. After repeated differences, low-degree polynomial dependence appears; eventually the determinant may vanish because the rows live in a space of too small dimension.

This is the general principle: if entries are values of a polynomial of degree less than the number of rows, finite differences can expose linear dependence.

§1.9 Triangularization by row operations

Several examples reduce a determinant to triangular form. One page shows a matrix whose first row is added to every other row; the determinant becomes upper triangular and the product of diagonal entries is obtained, for instance

$$D = n!.$$

Another example has a diagonal matrix plus a rank-one row of ones. If all diagonal entries a_i are nonzero, add suitable multiples of the other columns to the first column to get

$$D = \left(a_0 - \frac{1}{a_1} - \cdots - \frac{1}{a_n} \right) a_1 a_2 \cdots a_n.$$

If one of the a_i is zero, expansion along the corresponding row or column gives the exceptional formula. The source page explicitly separates the two cases.

§1.10 Recursive determinants

A determinant with ones on the diagonal or near the diagonal leads to a recurrence such as

$$D_n = D_{n-1} - D_{n-2} = -D_{n-3} = \cdots.$$

The page lists the resulting periodic cases depending on $n \bmod 3$. The key method is expansion along a sparse row or column after simplifying the matrix.

Another recursive example uses a tridiagonal determinant with diagonal $2a$ and off-diagonal a^2 and 1. Swapping rows or columns to make the matrix anti-diagonal creates a sign

$$(-1)^{n(n-1)/2},$$

which the handwritten note circles. This is a reminder that permutation signs are often the hidden danger in determinant shortcuts.

§1.11 Cramer's rule

The note then turns to linear systems. For the 2×2 system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2, \end{cases}$$

let

$$\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \Delta_1 = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}, \quad \Delta_2 = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}.$$

If $\Delta \neq 0$, there is a unique solution

$$x_1 = \frac{\Delta_1}{\Delta}, \quad x_2 = \frac{\Delta_2}{\Delta}.$$

If $\Delta = 0$ but not all numerator determinants vanish, the system has no solution. If all determinants vanish, there may be infinitely many solutions.

For an $n \times n$ system $Ax = b$, if

$$D = |A| \neq 0,$$

then

$$x_j = \frac{D_j}{D},$$

where D_j is obtained by replacing the j -th column of A by the right-hand-side vector b .

§1.12 Rank and minors

The later pages review rank. The rank of a matrix is the largest order of a nonzero minor:

$$r(A) = \max\{k : \text{extsomekeext-minor of } A \text{ extis nonzero}\}.$$

Equivalently, it is the dimension of the column space or row space.

The note states the standard consequences:

$$r(A) = r(A^T),$$

row and column elementary transformations preserve rank, and for square A ,

$$|A| \neq 0 \iff r(A) = n.$$

If $r(A) < n$, the columns are linearly dependent and the determinant vanishes.

§1.13 Solvability of linear systems

For a linear system

$$Ax = b,$$

let $[A|b]$ be the augmented matrix. The solvability criterion is

$$Ax = b \text{ extissolvable} \iff r(A) = r([A|b]).$$

If the common rank is r and the number of unknowns is n , then:

$$r = n \implies \text{extunique solution},$$

$$r < n \implies \text{extinfinitely many solutions}.$$

If

$$r(A) < r([A|b]),$$

then the system is inconsistent.

For the homogeneous system $Ax = 0$, the augmented rank is always the same as $r(A)$, so the system always has the zero solution. It has nonzero solutions iff

$$r(A) < n.$$

§1.14 Handwritten comments preserved

The note repeatedly warns against applying formulas mechanically. The important comments are:

- block determinant formulas require the zero block to be in the right place;
- determinant transformations are safe only if one tracks sign changes;
- row/column tricks are not independent recipes but attempts to create zeros, common factors, or known determinant patterns;
- Cramer's rule is clean but only useful when the coefficient determinant is nonzero;
- rank is the robust language for the singular cases where determinants vanish.

§1.15 The lesson behind it

This note is a determinant-method toolbox. Vandermonde handles power columns; block triangular form factors determinants; Laplace expansion exploits zeros; row and column transformations simplify structure; recursion evaluates patterned determinants; Cramer's rule solves nonsingular systems; rank handles singular systems. The conceptual progression is from explicit determinant computation to the structural question of linear dependence.

§1.16 Determinant as alternating multilinear invariant

The determinant is characterized by multilinearity, alternation, and normalization. This explains row operation rules, cofactor expansion, and why repeated rows give determinant zero.

§1.17 Vandermonde and evaluation maps

The Vandermonde determinant

$$\prod_{i < j} (x_j - x_i)$$

measures whether evaluation at distinct points gives independent conditions on polynomials. If two points coincide, two rows coincide and the determinant vanishes.

§1.18 Rank and linear systems

Rank measures the number of independent equations or independent output directions. A system is solvable exactly when the right-hand side lies in the column space.

§1.19 Determinants and rank as invariants

Determinant tricks, block forms, Vandermonde structures, and rank computations all track the same objects: volume scaling, independence, and solvability.

§1.20 Determinant through exterior powers

The determinant tricks in this note become more coherent through exterior algebra. A matrix $A : V \rightarrow V$ induces a map on the top exterior power:

$$\Lambda^n A : \Lambda^n V \rightarrow \Lambda^n V.$$

Since $\Lambda^n V$ is one-dimensional, this induced map is multiplication by a scalar. That scalar is $\det A$.

Laplace expansion, row operations, and block triangular formulas are all coordinate expressions of how A acts on oriented volume. For instance, if two columns become dependent, their wedge product is zero, so the determinant is zero.

§1.21 Schur complement

For a block matrix

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

with A invertible, block elimination gives

$$\det M = \det A \det(D - CA^{-1}B).$$

The matrix

$$D - CA^{-1}B$$

is the Schur complement of A in M .

This single formula explains many block determinant tricks. It also appears in Gaussian elimination, conditional covariance matrices, numerical linear algebra, and optimization.

§1.22 Matrix determinant lemma

A rank-one update has a particularly simple determinant formula:

$$\det(A + uv^T) = \det(A)(1 + v^T A^{-1}u)$$

when A is invertible. More generally,

$$\det(A + UV^T) = \det(A) \det(I + V^T A^{-1}U).$$

This is useful when a complicated determinant is a simple matrix plus a low-rank perturbation.

Many contest determinant problems secretly create low-rank updates by adding rows or columns. The advanced formula names the structure behind the trick.

§1.23 Rank varieties

The condition $\text{rank } A \leq r$ is algebraic: it is equivalent to the vanishing of all $(r+1) \times (r+1)$ minors. Therefore matrices of rank at most r form an algebraic variety in affine space.

For example, singular $n \times n$ matrices are the hypersurface

$$\det A = 0.$$

The elementary rank/minor criteria are thus equations defining geometric spaces. This is the doorway from linear algebra to algebraic geometry.

§1.24 Returning to Cramer and rank

Cramer's rule solves a nonsingular system by ratios of determinants. Rank criteria handle the singular cases. The advanced view says: the nonsingular matrices form the open set where the determinant volume form is nonzero; singular matrices lie on the determinantal hypersurface; rank strata describe how badly invertibility fails.

§1.25 Determinant identities as low-rank geometry

Many determinant problems are solved by adding all rows to one row, subtracting rows, or recognizing that a matrix has the form

$$D + uv^T.$$

The matrix determinant lemma says

$$\det(D + uv^T) = \det(D)(1 + v^T D^{-1}u)$$

when D is invertible. If D is diagonal, this turns a large determinant into one scalar correction term.

Geometrically, a rank-one update changes the image of the unit cube only in one coupled direction. Most independent volume scaling is still done by D ; the determinant lemma measures the one additional shear/stretch contribution.

§1.26 Schur complements and eliminating variables

Block Gaussian elimination is not merely a determinant shortcut. If

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} u \\ v \end{pmatrix}$$

and A is invertible, then the first block row gives $x = A^{-1}(u - By)$. Substituting into the second block row yields

$$(D - CA^{-1}B)y = v - CA^{-1}u.$$

The Schur complement is the effective system after eliminating x .

This interpretation connects determinant computation, solving linear systems, conditional covariance in probability, and constrained quadratic optimization.

§1.27 Determinantal varieties and singular loci

The rank condition $\text{rank } A \leq r$ defines a determinantal variety. Its equations are all $(r+1)$ -minors. The singular matrices form the hypersurface $\det A = 0$, and matrices of lower rank form deeper singular strata inside it.

Thus the elementary condition “determinant equals zero” is not merely a yes/no invertibility test. It defines a geometric boundary between invertible transformations and maps with collapsed directions. Cramer's rule works on the open complement of this hypersurface; rank methods describe what happens on the hypersurface.